Smart Cloud Forensics Evidence Detection Tool

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Digital forensics process in a cloud environment poses challenges in all stages of a typical forensics investigation. Most of the cyber-attacks in a cloud environment leave their traces on the cloud resources such as network, memory or disk. The data generated during such an incident can be a potential evidence, if it contributes information to prove or disprove an incident in question. All data generated in a cloud environment might not be evidence data, and there is lot of data that is generated in a cloud environment. A forensic investigation typically starts by collecting the data that is related to the incident. Detecting potential evidence data out of all the data generated in a cloud environment will save time as well as efforts of a forensics analyst.

We propose a tool, that monitors the virtual resource's activities smartly. The tool quantifies the activity at each resource, and predicts, if the data generated at time instance 't' is a potential evidence. The tools uses libvirt API to get monitoring data. The tool uses an AI Agent that triggers the evidence collection depending on the rate of activities at the virtual resource.

This tool also includes proposed new featureset to be considered for evidence classification. The tool has inbuilt Random Forest classifier programmed to classify the evidence. The inbuilt Random Forest has been programmed with extensive experimentation. However, the end user can always modify the parameters and train his own model with retrain module.

Researcher and academicians will find this tool useful to study cloud forensics in depth. The tool can be useful to generate one's own synthetic dataset, study correlation of attacks on the resource activities, live memory forensics etc. The tool can also cater to smart monitoring application in a Private Cloud Platform supported by libvirt.

The modules are explained in the documentation in details. The module placement and dependencies are explained in the Figure 1.

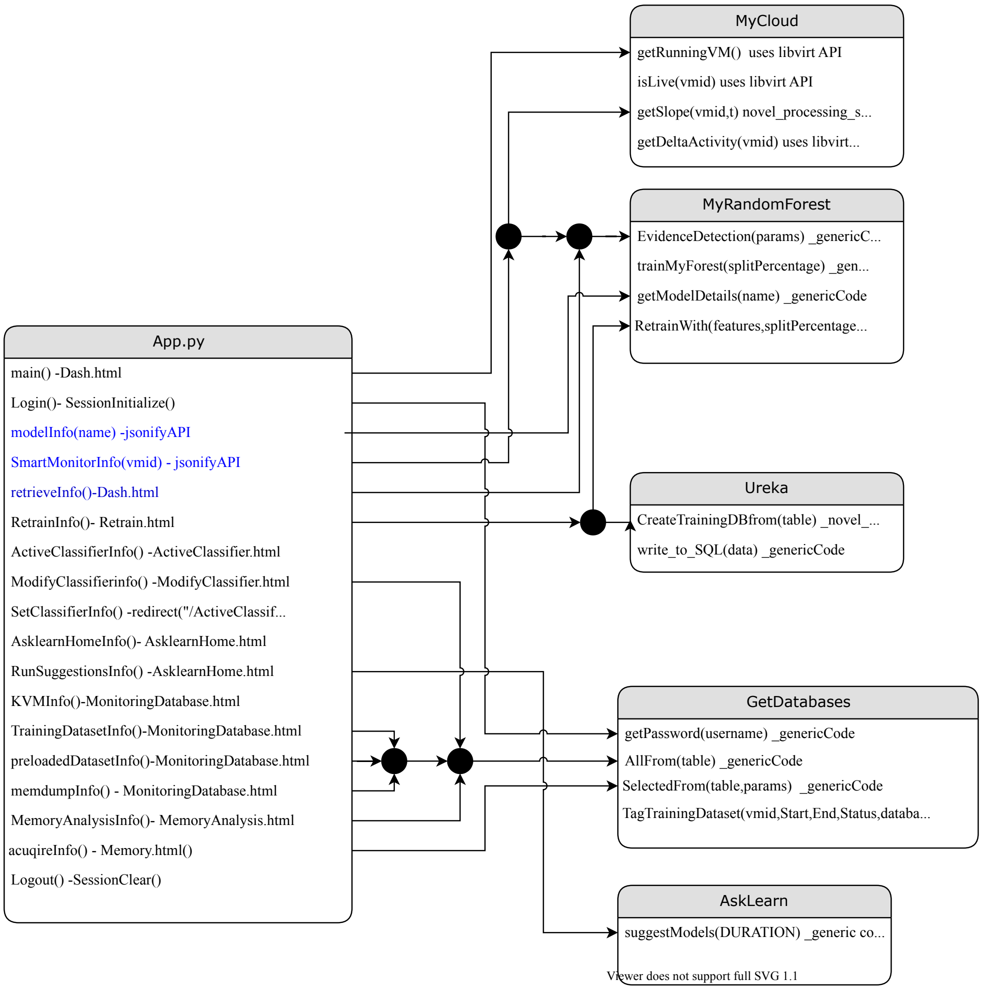

Figure 1: Smart Cloud Forensics Tool Components